

PE-EC702B — Digital Image and Video Processing

Semester VII • 3L:0T:0P • 3 Credits

Module 1 Exam Notes: Digital Image Fundamentals

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

Contents

1	Elements of Visual Perception	2
1.1	Eye Structure and Receptors	2
1.2	Contrast Discrimination and Weber Ratio	2
2	Image Sensing and Acquisition	2
3	Image Sampling and Quantization	3
3.1	Representing Digital Images	3
3.2	Resolution and Spatial Regressions	3
4	Basic Relationships Between Pixels	3
4.1	Pixel Neighborhoods	4
4.2	Adjacency and Path Connectivity	4
4.3	Distance Measures	4
5	Worked Examples	5

1. Elements of Visual Perception

Understanding human visual perception is crucial for digital image processing because the human eye is the ultimate receiver of the visual output.

Visual Perception Fundamentals

Brightness Adaptation: The human eye can operate over an enormous dynamic range (10^{10} units), but cannot view this entire range simultaneously. It accomplishes this by changing its overall sensitivity, a process known as *brightness adaptation*.

1.1 Eye Structure and Receptors

The human eye contains two primary photoreceptors located on the retina:

- **Cones (Photopic Vision):** Approximately 6 to 7 million receptors located mainly in the central portion of the retina (fovea). Highly sensitive to color. Allows resolve of fine details.
- **Rods (Scotopic Vision):** Approximately 75 to 150 million receptors distributed over the retinal surface. Give a general, overall picture of the field of view. Sensitive to low illumination levels (night vision), but are color-blind.

1.2 Contrast Discrimination and Weber Ratio

The ability of the eye to discriminate between different intensity levels is characterized by the **Weber Ratio**.

- A subject is exposed to a uniform background intensity I .
- A short duration increment ΔI is added.
- The ratio $\frac{\Delta I}{I}$ at which the subject can just detect the change is called the **Weber Fraction**.
- A small Weber fraction indicates good intensity discrimination (high sensitivity), whereas a large fraction indicates poor discrimination (occurring at low illumination).

2. Image Sensing and Acquisition

To generate a digital image, the physical energy (light, X-ray, electromagnetic wave) must be converted into electrical signals by sensors.

Image Sensor Configurations

Image Sensors: Devices that convert illumination energy into electrical voltage. There are three primary configurations:

- (a) **Single Sensor:** Uses a photodiode that moves mechanically in x - and y -directions (e.g., microdensitometers).
- (b) **Sensor Strips:** Inline array of sensors providing 1D line scans (e.g., flatbed scanners).

- (c) **Sensor Array:** 2D grid of sensors capturing the entire scene (e.g., CCD/CMOS arrays in digital cameras).

3. Image Sampling and Quantization

A continuous image $f(x, y)$ must be converted into digital format in both spatial coordinates (sampling) and amplitude (quantization).

Sampling vs. Quantization

Sampling: Digitizing the continuous spatial coordinates (x, y) into discrete grid indices.
Quantization: Digitizing the continuous amplitude (intensity values) into discrete gray levels.

3.1 Representing Digital Images

A digital image of size $M \times N$ with L discrete gray levels is represented as a matrix:

$$f(x, y) = \begin{bmatrix} f(0, 0) & f(0, 1) & \cdots & f(0, N-1) \\ f(1, 0) & f(1, 1) & \cdots & f(1, N-1) \\ \vdots & \vdots & \ddots & \vdots \\ f(M-1, 0) & f(M-1, 1) & \cdots & f(M-1, N-1) \end{bmatrix} \quad (1)$$

The number of gray levels L is typically a power of 2:

$$L = 2^k \quad (2)$$

where k is the number of bits required to store each pixel. The total number of bits b required to store the image is:

$$b = M \times N \times k \quad (3)$$

3.2 Resolution and Spatial Regressions

- **Spatial Resolution:** The smallest discernible detail in an image, measured in dots/pixels per unit distance (e.g., dpi). Insufficient spatial sampling causes **aliasing** (moiré patterns).
- **Intensity Resolution:** The smallest discernible change in gray level, measured in bits per pixel.
- **False Contouring:** An effect caused by insufficient quantization levels (typically $k < 4$ bits), producing visible, artificial boundaries between smooth gray-level transitions.

4. Basic Relationships Between Pixels

Let p and q be pixels in an image. Their relationships determine boundaries, regions, and structures.

4.1 Pixel Neighborhoods

- **4-neighbors** ($N_4(p)$): The 4 horizontal and vertical pixels adjacent to p at coordinates:

$$(x+1, y), \quad (x-1, y), \quad (x, y+1), \quad (x, y-1) \quad (4)$$

- **Diagonal neighbors** ($N_D(p)$): The 4 diagonal pixels adjacent to p at coordinates:

$$(x+1, y+1), \quad (x+1, y-1), \quad (x-1, y+1), \quad (x-1, y-1) \quad (5)$$

- **8-neighbors** ($N_8(p)$): The union of 4-neighbors and diagonal neighbors:

$$N_8(p) = N_4(p) \cup N_D(p) \quad (6)$$

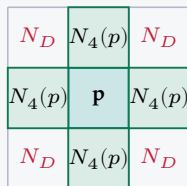
4.2 Adjacency and Path Connectivity

Let V be the set of intensity values used to define adjacency (e.g., $V = \{1\}$ in binary images, or $V = \{128, \dots, 255\}$ in grayscale).

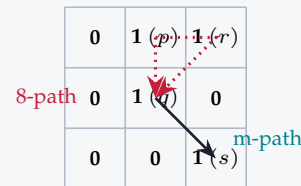
1. **4-adjacency**: Two pixels p and q with values in V are 4-adjacent if $q \in N_4(p)$.
2. **8-adjacency**: Two pixels p and q with values in V are 8-adjacent if $q \in N_8(p)$.
3. **m-adjacency (mixed adjacency)**: Two pixels p and q with values in V are m-adjacent if:
 - $q \in N_4(p)$, OR
 - $q \in N_D(p)$ and the intersection $N_4(p) \cap N_4(q)$ has no pixels with values from V .

Purpose: m-adjacency is introduced to eliminate the multiple path ambiguities that arise in 8-adjacency.

Pixel Neighborhoods and m-Adjacency



Pixel Neighbors



m-path resolves 8-path ambiguity

4.3 Distance Measures

Let $p(x, y)$, $q(s, t)$, and $z(v, w)$ be pixels. A distance function $D(p, q)$ must satisfy:

- $D(p, q) \geq 0$ ($D(p, q) = 0$ if and only if $p = q$)
- $D(p, q) = D(q, p)$ (Symmetry)
- $D(p, z) \leq D(p, q) + D(q, z)$ (Triangle Inequality)

Key Distance Metrics**Euclidean Distance (D_e):**

$$D_e(p, q) = \sqrt{(x - s)^2 + (y - t)^2} \quad (7)$$

City-Block Distance (D_4):

$$D_4(p, q) = |x - s| + |y - t| \quad (8)$$

Chessboard Distance (D_8):

$$D_8(p, q) = \max(|x - s|, |y - t|) \quad (9)$$

5. Worked Examples**Distance Calculations and Path Analysis**

Problem: Given pixel p at coordinates $(3, 4)$ and pixel q at coordinates $(7, 8)$ in a digital image:

- Calculate the Euclidean distance $D_e(p, q)$.
- Calculate the City-Block distance $D_4(p, q)$.
- Calculate the Chessboard distance $D_8(p, q)$.
- Determine if there is an m-path between p and q using $V = \{1\}$ in the following 3×3 grid segment, where the coordinates are:

$$a(0, 2) = 0 \quad b(1, 2) = 1 \quad c(2, 2) = 1$$

$$d(0, 1) = 0 \quad e(1, 1) = 1 \quad f(2, 1) = 0$$

$$g(0, 0) = 0 \quad h(1, 0) = 0 \quad i(2, 0) = 1$$

Solution:**(a) Euclidean Distance (D_e):**

$$D_e(p, q) = \sqrt{(3 - 7)^2 + (4 - 8)^2} = \sqrt{(-4)^2 + (-4)^2} = \sqrt{16 + 16} = \sqrt{32} \approx 5.66$$

(b) City-Block Distance (D_4):

$$D_4(p, q) = |3 - 7| + |4 - 8| = 4 + 4 = 8$$

(c) Chessboard Distance (D_8):

$$D_8(p, q) = \max(|3 - 7|, |4 - 8|) = \max(4, 4) = 4$$

(d) m-path Analysis:

- We seek a path of pixels with value 1 between $e(1, 1)$ and $i(2, 0)$ under m-adjacency rules.

- Let's check neighbors of $e(1, 1)$:
 - $b(1, 2)$ has value 1 and is a 4-neighbor of e ($b \in N_4(e)$). Thus, there is an m-path segment $e \leftrightarrow b$.
 - $i(2, 0)$ is a diagonal neighbor of e ($i \in N_D(e)$).
 - Under m-adjacency rules, diagonal pixels e and i are m-adjacent only if the intersection of their 4-neighbors has no pixels of value $V = \{1\}$.
 - $N_4(e) \cap N_4(i) = \{(2, 1), (1, 0)\} = \{f, h\}$.
 - Since $f(2, 1) = 0$ and $h(1, 0) = 0$, both intersection pixels have value 0, which is not in V .
 - Therefore, e and i are directly m-adjacent. The path is $e \leftrightarrow i$ (length 1).
- Now let's check path from $b(1, 2)$ to $i(2, 0)$:
 - $c(2, 2)$ is in $N_4(b)$ and has value 1. Thus, $b \leftrightarrow c$.
 - $i(2, 0)$ is a diagonal neighbor of f , but c and i are not adjacent. The only valid path is $b \leftrightarrow e \leftrightarrow i$.

Quick Revision — Module I

Digital Image Fundamentals | PE-EC702B

Key Formulas at a Glance

Quantization Levels	$L = 2^k$ (k = bit depth)
Storage Requirement	$b = M \times N \times k$ (bits for an $M \times N$ image)
Euclidean Distance	$D_e(p, q) = \sqrt{(x - s)^2 + (y - t)^2}$
City-Block Distance	$D_4(p, q) = x - s + y - t $
Chessboard Distance	$D_8(p, q) = \max(x - s , y - t)$
Weber Fraction	Weber Ratio = $\frac{\Delta I}{I}$ (ΔI = just noticeable increment)

One-Line Definitions

- **Photopic Vision:** Vision mediated by cone cells under bright light, providing high color and spatial resolution.
- **Scotopic Vision:** Vision mediated by rod cells under low light, providing color-blind, coarse spatial vision.
- **Sampling:** Discretizing continuous coordinates (x, y) to define spatial resolution.
- **Quantization:** Discretizing continuous amplitude (gray level) to define intensity resolution.
- **False Contouring:** Artificial banding appearing in regions of smooth transitions due to low quantization.
- **m-Adjacency:** Mixed adjacency that prevents dual-path ambiguities by checking vertical/horizontal neighbors.

Frequently Asked / Viva Questions

1. State the difference between rods and cones.

Cones are located mainly in the fovea, are color-sensitive, and operate in bright light (photopic vision). Rods are distributed across the retina, are highly sensitive to low light (scotopic vision), and are color-blind.

2. What is Weber ratio? What does a low Weber ratio imply?

It is the ratio of the just noticeable increment in intensity ΔI to the background intensity I . A low Weber ratio implies high intensity discrimination capability of the human eye.

3. Why is m-adjacency preferred over 8-adjacency?

8-adjacency can lead to multiple, ambiguous paths between two pixels (e.g., both diagonal and horizontal-vertical paths simultaneously). m-adjacency resolves this by choosing the diagonal path only when no horizontal-vertical path exists.

Frequently Asked / Viva Questions (contd.)

4. Explain the concept of false contouring.

If the number of quantization levels L is too low (typically < 16 levels), the smooth transitions of intensity appear as discrete, nested bands of constant gray level with sharp, artificial boundaries.

5. How does city-block distance differ from chessboard distance?

D_4 distance (city-block) allows only horizontal and vertical movements, while D_8 distance (chessboard) allows both horizontal-vertical and diagonal movements. For any two points, $D_8(p, q) \leq D_4(p, q)$.

6. What is spatial resolution and how is it measured?

It is the smallest discernible detail in an image. It is measured in terms of line pairs per unit distance or dots per inch (dpi).

7. What is the storage size of a 512×512 image with 256 gray levels?

$256 = 2^8$, so $k = 8$ bits. Total size $b = 512 \times 512 \times 8 = 2,097,152$ bits = 256 KB.

8. What causes aliasing in digital images?

Aliasing occurs when the sampling frequency is less than twice the maximum spatial frequency of the image (violating the Nyquist criterion), causing high-frequency details to wrap as moiré patterns.

9. Define a path between two pixels p and q .

A path is a sequence of distinct pixels $(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$ where $(x_0, y_0) = p$, $(x_n, y_n) = q$, and (x_i, y_i) is adjacent to (x_{i-1}, y_{i-1}) for all $1 \leq i \leq n$.

10. What are the components of an image sensing device?

An image sensing device consists of a filter (to select wavelength band), a sensing material (to convert photon energy to charge), and a readout circuit (to convert charge to voltage).

Comparison of Pixel Distances

Distance Metric	Path allowed	Value for $p(0, 0)$ and $q(2, 2)$
Euclidean (D_e)	Straight line (any angle)	$\sqrt{2^2 + 2^2} = \sqrt{8} \approx 2.83$
City-Block (D_4)	Horizontal and vertical only	$2 + 2 = 4$
Chessboard (D_8)	Horizontal, vertical, and diagonal	$\max(2, 2) = 2$